



Plant Disease Classification Using Computer Vision

NAME: SAIRA JABEEN

ROLL NO: 23-AI-10

SECTION: A2(B)

SEMESTER: 6TH

YEAR: 3RD

**DEPARTMENT: ARTIFICIAL
INTELLIGENCE**

COURSE: MACHINE LEARNING

COURSE INSTRUCTOR: ENGR.

HAMZA FAROOQI

1. Introduction

Plant diseases significantly affect agricultural productivity and crop quality. Early detection of plant diseases can help farmers take preventive measures and reduce losses.

The objective of this project is to develop an automated plant disease classification system using Computer Vision and Deep Learning techniques. The PlantVillage dataset was utilized to classify plant diseases from leaf images.

2. Dataset Description

The PlantVillage dataset contains thousands of leaf images belonging to multiple healthy and diseased plant classes.

Characteristics of the dataset:

- RGB color images
- Multiple disease categories
- Different plant species
- Balanced image samples

The dataset was downloaded from Kaggle and organized into class-wise folders.

3. Image Preprocessing

To improve image quality and reduce noise, several preprocessing techniques were applied.

3.1 Image Resizing

All images were resized to:

64×64 pixels

This reduced computational cost while maintaining important visual information.

3.2 Grayscale Conversion

RGB images were converted to grayscale to simplify processing and emphasize texture information.

3.3 Histogram Equalization

Histogram Equalization was applied to enhance image contrast and improve visibility of disease symptoms.

3.4 Normalization

Pixel values were normalized between 0 and 1 for faster and more stable model training.

4. Feature Extraction Using Computer Vision

Edge detection techniques were used to highlight important leaf structures.

Sobel Filter

The Sobel operator was applied in both X and Y directions to detect horizontal and vertical edges.

Laplacian Filter

The Laplacian operator was used to detect edges in all directions and emphasize disease-related patterns.

These techniques helped visualize texture and boundary information in plant leaves.

5. Data Preparation

Images were loaded class-wise and assigned numerical labels.

The dataset was divided into:

- Training Set: 80%
- Testing Set: 20%

This allowed evaluation of model performance on unseen data.

6. Deep Learning Models

6.1 Fully Connected Neural Network (FCNN)

A basic neural network was developed consisting of:

- Flatten Layer
- Dense Layer (128 neurons)
- Dense Layer (64 neurons)
- Softmax Output Layer

The model was trained using:

- Adam Optimizer
 - Sparse Categorical Crossentropy Loss
 - Accuracy Metric
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6.2 Transfer Learning Using MobileNetV2

To improve classification performance, MobileNetV2 was used as a pretrained feature extractor.

Advantages

- Faster convergence
- Better feature representation
- Reduced training time
- Higher classification accuracy

The final architecture included:

- MobileNetV2 Base Model
 - Global Average Pooling Layer
 - Dense Layer
 - Dropout Layer
 - Softmax Classification Layer
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7. Fine-Tuning

The last layers of MobileNetV2 were unfrozen and retrained.

Fine-tuning allowed the network to learn domain-specific plant disease features and further improve performance.

8. Performance Evaluation

The trained models were evaluated using classification accuracy.

Observations:

- FCNN successfully learned disease patterns from images.

- MobileNetV2 outperformed the FCNN model.
 - Transfer learning provided better generalization and feature extraction capabilities.
 - Fine-tuning further enhanced model performance.
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9. Challenges and Solutions

Challenge	Solution
Large image dataset	Resized images
Uneven lighting conditions	Histogram Equalization
High-dimensional image data	Feature extraction through CNN
Limited training efficiency	Transfer Learning
Overfitting risk	Dropout and Fine-Tuning

10. Conclusion

This project successfully implemented a plant disease classification system using Computer Vision and Deep Learning techniques. Image preprocessing and edge detection enhanced important visual features, while MobileNetV2 transfer learning achieved superior classification performance.

The project demonstrates how modern computer vision techniques can support smart agriculture by enabling automated and accurate plant disease detection.

Technologies Used

- Python
- OpenCV
- TensorFlow
- Keras
- NumPy
- Matplotlib
- Scikit-Learn

Models Used

1. Fully Connected Neural Network (FCNN)
2. MobileNetV2 Transfer Learning Model

